Paolo De Marinis

MSc in Mathematics | Technical & Scientific Projects | Cryptography, Data & Linux

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Professional Profile

MSc in Mathematics (106/110) with experience in technical and scientific projects spanning zk-STARKs and cryptography, reversible geospatial data compression, quantitative analysis, Linux/ACPI diagnostics and C programming. My Master's thesis, written in English, is a largely self-contained theoretical reconstruction of zk-STARK protocols with formal contributions on Algebraic Intermediate Representations (AIR) and their optimization. I have also undertaken structured independent study in cryptography and DLT. Direct programming and scientific-computing experience is strongest in C and MATLAB; Python experience is project-based, centered on pipeline structure and design, code reading and review, targeted modifications, testing and validation. I use agentic AI workflows for implementation and technical analysis support while retaining responsibility for objectives, constraints, review, testing, validation and final acceptance.

Education

MSc in Mathematics (LM-40) - 106/110 2023-2025

University of Insubria, Italy

Thesis: *Fast Fourier Transform in zk-STARK Protocols: Interpolation, Encoding and Proximity Proofs*, written in English. Largely self-contained theoretical reconstruction of zk-STARKs from first principles. Theoretical contributions include a formal derivation from a finite-state machine to a Computational Integrity statement and Algebraic Intermediate Representation (AIR), with proof of equivalence between AIR satisfiability and CI-statement validity; a general AIR definition unifying formulations in the literature; and an optimization framework based on two preorders for valid and probabilistically sound extensions. The thesis also reconstructs the STARK protocol through Reed-Solomon encoding, DEEP and FRI, with complexity and soundness analysis, and covers the hash-based commitment structure and Merkle trees used within the zk-STARK construction.

Technical Skills

- **Cryptography and DLT:** studied foundations and security models of modern cryptography, symmetric cryptography, hash functions and MACs, core public-key topics including RSA and Diffie-Hellman, ECDSA and Schnorr digital signatures, and the foundations of Bitcoin, Ethereum and DLT. Continuing study includes additional public-key topics, PKI/TLS, zero-knowledge beyond zk-STARKs, advanced DLT topics, privacy-enhancing technologies and post-quantum cryptography.
- **Programming and scientific computing:** direct C and MATLAB experience. University C++ work was mainly procedural and C-style; object-oriented C++ was limited to a single program for the pass/fail Programming Languages exam. GNU Octave was used directly in the Numerical Analysis exam; Maple was learned independently and used for orbital-transfer and rendezvous calculations presented in the Space Mechanics examination seminar. University training in algorithms and data structures.
- **Python project work:** pipeline structure and design, code reading/review, targeted changes, testing and validation for a reversible pipeline mapping 2D points to Hilbert indices, ordering the indices and applying delta encoding to the ordered sequence, with inverse decoding from deltas to indices and from indices back to points. The implementation used `hilbertcurve`, with targeted ChatGPT support for some functions.
- **Agentic AI workflows:** objectives and constraints definition, context preparation, diff review, test execution and monitoring, result validation and final acceptance. Current agentic development primarily uses Codex and Claude Code/Cowork; technical analysis primarily uses ChatGPT and Claude. Additional experience includes Gemini, Cursor and Google Antigravity.
- **Technical environment and tools:** Linux daily since 2012; practical use of Git/GitHub and GitHub Actions; advanced LaTeX for academic and technical documentation; project-specific experience with ACPI/AML diagnostics and validation, initramfs and Limine through OMEN ACPI Toolkit. Limited direct project experience with HTML and SQL.

Selected Projects

OMEN ACPI Toolkit - open-source personal project Jul 2026-present (maintenance)

Repository | v2.4.0 release (14 Aug 2026); unaffiliated with HP. On the sole physically validated configuration—HP OMEN MAX 16-ap0006sl, board 8E35, BIOS F.13—used logs and controlled experiments to distinguish incomplete S5 shutdown, which left the dGPU powered, from a separate WQBZ ACPI error. Defined an S5-scoped override that sets `PEGP.OMPR=3` and invokes `PEGP._PS3()` from `_PTS(5)`, using an existing firmware path. I own the hardware evidence, objectives and constraints, safety decisions, diff review, log/regression analysis, real-hardware validation and releases. Codex made substantial contributions to the Bash/Python implementation, automated tests and documentation; no independent specialist or line-by-line audit is claimed. The toolkit includes local ACPI-table capture, round-trip checks, fail-closed guards, recovery tooling and separate Limine entries that preserve stock boot paths. v2.4.0 refactors packaging, documentation and release engineering; ACPI transformers, verifiers, patches and fixtures are unchanged from the v2.3.0 validated baseline.

Geospatial data compression - algorithmic contribution 2023

Voluntary, unpaid scientific collaboration; non-author contribution acknowledged in Liverotti et al., *International Journal for Numerical Methods in Engineering* (2025). Contributed to the design and Python implementation of a reversible pipeline for 2D data: mapping points to Hilbert indices, ordering by those indices, applying delta encoding to the ordered sequence, decoding the deltas, and reconstructing the original points through the inverse Hilbert map. Personally verified round-trip reversibility on real Earth Observation data. The implementation used `hilbertcurve`, with targeted ChatGPT support for some functions. Also proposed Huffman coding for brightness values and contributed to the triangulation-reconstruction approach.

RIVES cartridges - C games and procedural geometry 2024; GitHub repositories 2026
Original production and Jam (Oct–Dec 2024): conceived, developed and published *Bomb Flip* first and then *Slither Slide* in C with RIVES (`riv.h`). *Bomb Flip* is inspired by Voltorb Flip, not a clone: I designed its rules, scoring, timer, scanner, Fold action and progression, and wrote most original gameplay code; Cursor mainly supported animation and implementation details. For *Slither Slide*, I designed mechanics, worked on parts of the code, integrated and validated the result, and selected/configured the closed uniform cubic B-spline; assistance was greater. *Slither Slide* placed 2nd out of 7 entries in RIVES Jam #3. GitHub repository phase (Aug 2026): I created and now maintain the public *Bomb Flip* (v1.0.0) and *Slither Slide* (v1.0.0) repositories from the available post-jam versions. They are not exact archival snapshots of the 2024 source. I organized code and provenance and integrated documentation, tests, CI, `.sqfs` cartridges and checksums; Codex assisted with maintenance, refactoring, tests and documentation, which I reviewed, integrated and validated.

Coffee extraction - quantitative analysis in MATLAB Aug 2019-Feb 2020
Independent voluntary and unpaid project requested by Caffè dei Cavalieri, Pisa. Designed the analysis and calculations, measured total dissolved solids (TDS), built least-squares extraction curves in MATLAB and interpreted the results for a mixed infusion/percolation process. The analysis was presented by Matteo Cecconi at the 2020 Italian Brewers Cup, where he placed 2nd.

Blockchain & Community Background

Practical interest in blockchain began in 2016 and resumed through structured self-study and hands-on use from 2020. Experience includes Bitcoin and Ethereum self-custody with Ledger and MetaMask, dApps, DeFi, wallet-based transaction signing, smart-contract interaction and initial experimentation with Solidity. In 2022, the Cartesi Foundation published the community profile *Why a School Teacher Chose to Hold CTSI*, covering my path from Ethereum and DeFi to the scalability problem and interest in Cartesi. The profile reflects community participation and personal study, not employment by Cartesi.

Additional Education

Master's studies in Mathematics - 75 ECTS completed before transfer 2016-2023
University of Pisa, Italy
Completed 75 ECTS through October 2020; enrollment was subsequently maintained during a study pause before transfer to the University of Insubria in 2023. Of the 75 ECTS, 71 were recognized on transfer. Coursework included Numerical Analysis, Scientific Computing, Dynamical Systems, Orbital Determination, Celestial Mechanics, Space Mechanics and Planetary Systems. Academic work included direct use of GNU Octave in the Numerical Analysis exam; use of independently learned Maple for orbital-transfer and rendezvous calculations presented in the Space Mechanics examination seminar; preparation of an independent 47-slide seminar on Gauss, Laplace and Mossotti methods for preliminary orbit determination; and an individual study of the Yarkovsky effect on asteroids for Planetary Systems.

BSc in Mathematics (L-35) 2012-2016
Sapienza University of Rome, Italy
Coursework included procedural, C-style programming in C++, algorithms and data structures, and MATLAB. Thesis on Brouwer's fixed-point theorem and periodic solutions of ODEs, including numerical phase portraits in MATLAB.

Professional Experience

Computer Science Teacher (A-41) & Digital Trainer Sep 2023-Aug 2024
IIS Leonardo da Vinci and Istituto Professionale Crotto Caurga, Italy
Taught computer science across technical and vocational programs, including C/C++ programming, spreadsheets and digital tools. Served on the digital team and delivered adult training on Google Workspace, plus EIPASS-related courses.

Mathematics Teacher Sep 2022-Aug 2023
IIS Agrario Antonio della Lucia, Feltre, Italy
Taught mathematics across technical, vocational and professional-training programs.

Additional Technical Experience

PrestaShop e-commerce customization - voluntary, unpaid May 2016-2022
New Zealand Sport S.r.l.s.
Set up and maintained aorakiski.com from an existing platform and theme; directly performed targeted HTML customization, module configuration and SQL queries and made targeted changes to data and search-related settings.

Languages

Italian: native. English: B2; used to write the MSc thesis and technical documentation and to read scientific literature.